

MAKERERE UNIVERSITY
MAKERERE UNIVERSITY BUSINESS SCHOOL
FINAL EXAMINATION FOR THE DEGREE OF BACHELOR OF BUSINESS
COMPUTING OF MAKERERE UNIVERSITY
ACADEMIC YEAR 2024-2025

Mission 85 and above
Course: *MAIRO STEPHEN (De siew)* Project Research Methods
Year of Study: II Semester: II
Time: 9am-12pm

Course Code: BUC3111
Date: May 05 2025
Duration: 3 Hours

Instructions to candidates:

- a) Answer any four (04) questions of your choice
- b) Do not write on the question paper
- c) Find other instructions on the answer booklet

Question 1:

(25 Marks)

- a) Explain the difference between research values and ethics. (05 Marks)
- b) What is the rationale for conducting scientific research? (10 Marks)
- c) Using examples from computer science and IT projects, describe five (05) the different types of research. (10 Marks)

Question 2:

(25 Marks)

A researcher team from the Bachelor of Business Computing Year II class is developing a Machine-learning soil testing system for Lutembe Integrated firm in Entebbe Municipality. In order to ensure the credibility of the project, they decide to use established principles of scientific research.

- a) Draft a problem statement for this project. *Real Scenario* (04 marks)
- b) Explain the importance of objectivity, originality, and informed consent in ensuring a high-quality research project. *the ability to make decisions* (06 Marks)
- c) What literature sources can the team use to get more insights about machine learning systems, before they start writing their proposal? *the consequences proposed research* (08 Marks)
- d) Explain the stages of literature review that the team will go through. (07 marks)

Question 3:

(25 Marks)

Julian, a student of Bachelor of Office and Information Management of MUBS is conducting research project to develop an integrated system for customer data security assurance in a financial institution. During requirements collection, the student is offered sensitive customer information by an employee who asks for a small "appreciation fee" in return.

- a) With examples, explain the kind of system requirements that the student is going to collect. (08 marks)
- b) Identify and discuss the ethical issues raised in this situation. Show how the student can ensure ethical compliance on her project. (06 Marks)

- c) Suggest the most appropriate Research Design that Julian can use. Basing on your choice, use an illustration to describe the research process that the student is going to undertake to deliver the project. (11 marks)

Question 4:

(25 Marks)

A student team is writing a research proposal on developing an e-commerce site for SMEs in Jinja City, in Uganda.

- a) Develop three research objectives and three corresponding research questions for this study. (09 Marks)
- b) Identify and write 5 appropriate assumptions that are necessary for this project to be implemented. (10 marks)
- c) Summarise the systems analysis processes that the team is going to undertake during project design. (06 marks)

Question 5:

(25 Marks)

Following the recent story by NTV Uganda and the Daily Monitor about drug abuse among university students, a group of undergraduate researchers is developing a project to combat this problem. The team wants develop a Mobile Interface, meant to be integrated in WhatsApp and other social media applications. The aim of the project is to sensitize MUBS students about the dangers of drug abuse, and provide psychosocial help to victims. The team needs to determine an appropriate research design, develop a project design plan, and collect requirements relating to this project.

- a) Identify, recommend, and illustrate the most appropriate system development approach for this project. Give reasons for your recommendation. (10 Marks)
- b) Identify and explain the qualitative and quantitative data collection techniques that would be suitable for this project. (15 Marks)

Question 6:

Write short and relevant notes about 5 of the following key research concepts. Insert some examples where necessary (25 Marks)

- a) Research variables (05 Marks)
- b) Cross-sectional research designs (05 marks)
- c) Literature review and its significance in research (05 Marks)
- d) APA referencing and its role in academic writing (05 Marks)
- e) Plagiarism and how to avoid it (05 Marks)
- e) The rationale for sampling in research (05 marks)

.....End of Examination.....